

ANNEX K:

CLOUD BUSINESS GUIDANCE

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Annex K: Cloud Business Guidance

1.0 INTRODUCTION

This Annex provides NGA's guidance on business practices in support of transitioning and cloud-native development activities. The Annex is a collection of specific business topics that are intended to provide subject context, implementation procedures and anticipated outcomes for cloud business operations. These business topics are identified in the following Table.

Section	Business Topic
2.0	Cloud Budgeting Practices
3.0	Cloud Financial Planning
4.0	Enterprise Service Authorization (and Lifecycle)
5.0	Cloud Environment Provisioning
6.0	Cloud Billing and Monitoring
7.0	Activity Optimization
8.0	Cloud Software Lifecycle and Licensing

2.0 CLOUD BUDGETING PRACTICES

This section provides an overview of NGA's budgeting and financial management practices for a cloud environment. It identifies the methods how NGA will operate and manage cloud-based Information Technology (IT) from a business perspective.

The key cloud operational objectives are to increase analytic performance and realize reductions in overall infrastructure costs.¹ When designed and implemented properly, a cloud computing business model should effect:

- Improved IT resource management through on demand Cloud Service Provider (CSP) services and performance metrics
- Increased focus on geospatial mission services via IT infrastructure economies
- Better NGA financial data tracking and Future Years Defense Program (FYDP) planning
- Better IT operations management

Efficient financial management of cloud-based services necessitates a commitment to greater transparency in the budgeting and allocation of IT resources. The quest for fiscal IT efficiencies reinforces the need to adjust to the cloud business model. NGA will apply the accounting processes associated with IC ITE capabilities for generating GEOINT services. As NGA incorporates the CSP services, funding process adjustments may be required. Program Managers (PMs) and cross-functional budgeting teams will assess and recommend business model changes (e.g., acquisition, contracting, and budgeting) for GEOINT services. The PMs and the Financial Management (FM) Directorate will adjudicate the budget authorizations and the Full Time Equivalent (FTE) contractor labor expenditure allocations by program.

2.1 GUIDANCE

Guidance: *The NGA Program Build should clearly identify cloud development and operations costs.*

¹ DNI Memorandum E/S 00564, [Guiding Principles for Implementing and Operating in a Common Intelligence Community Information Technology Enterprise](#)

Guidance: *Program Managers shall maintain cloud performance and cost data for the program and costs will be readily available to Financial Management (FM).*

The Program Build should provide transparency into the allocation of NGA funds and should include cloud maintainability and sustainment resource requirements. The Program Build supports NGA missions aligned with defined FYDP goals and is composed of a sequence of milestones and fiscal planning events (see [Appendix A - Program Build Milestones and Events](#)). The Resource & Capabilities Working Group provides a list of investments for the current year President's Budget baseline, approved by the Director, along with approved offsets (divestments) to assure compliance with oversight guidance and controls.

The Cloud PMO and CUO teams provide detailed cost and resource utilization data reports for each program to show utilization and data consumption for PMs to review and make resource adjustments. The PMs are required to provide data on cloud usage and costs to the FM Directorate. This data provides the necessary fiscal data fidelity, at the activity level to ensure decision makers have the necessary information to make informed budget decisions.

Guidance: *If the program needs additional funding in the year of execution, the PM should prepare a business case and an Unfunded Requirement (UFR) for leadership to adjudicate.*

If the Directorate cannot take care of the need; then, submit the business case and UFR for a leadership decision, on a priority basis, that need against other corporate UFRs.

Guidance: *FM should issue programming guidance and aid in resolving funding shortfalls via UFR approval and follow up once a UFR is approved by the program's directorate or leadership.*

The PM, in coordination with their FM representative, should maintain a structured process for identifying and the allocation of funds for existing and planned development and operations costs, to include cloud migration and operating costs. Cloud resource planning should be derived from the following estimating methods: analogy, parametric, engineering, and actual costs. The FM is to ensure that funding is available for both cloud development and sustainment in the out-years.

Guidance: *Program Managers shall be responsible for the management of the program's cloud development and operations costs.*

The PM shall consider the following cost measures for managing the cloud budgeting process:

- C2S billing for each program
- Managing computing and storage costs (i.e., usage of C2S tools)
- Resources needed for cloud transition and operating costs
- Resources needed to build cloud portability
- Resources needed to decompose data into decoupled data services
- Program license needs via enterprise environment
- Dependencies to other programs and services, including other agencies
- Scalability of services for mission and delivery environment

NGA via the CSP, provides detailed cost and resource utilization data for each program to show utilization and data consumption for use by programs to forecast future resource needs and to make necessary adjustments.

Performance metrics are to be used to detect problems, support decisions, and gauge the health of the program. Programs have ready access to system performance data to enable the PM to proactively manage cloud expenditures, through informed cloud service adjustments.

Performance metrics are a critical component to the program's business management. These metrics include information such as distinct user consumption/need, capacity, and resource utilization. CIO-T uses metrics in its cloud resource and operations support, as appropriate.

Guidance: *The Program Budget Review (PBR) stakeholders shall have oversight responsibility of the mission alignment and budgeting of the resources for IT programs.*

The PBR goal for the Program Build is to ensure a clear and transparent process for ensuring the best alignment of resources to achieve GEOINT services and mission support. The Advanced, Technical, Leadership, Analytical Suite (ATLAS) [FM View](#) contains the PBR Tool for documenting the invest, divest, and realignment throughout the program build process. The cloud costs include all NGA cloud transition and operations costs, as well as, the integration of the cloud environment across the National System for Geospatial Intelligence (NSG), the DoD Joint Information Environment (JIE) and IC Information Technology Enterprise (IC ITE).

Guidance: *The current IT business structure shall continue; including the associated processing of receipts, distribution of funding, and reports on contract performance and spending rates.*

NGA uses established processes to manage the business and financial aspects of IC ITE services and reporting. NGA applies the budget-based funding model, supported by the Intelligence Planning, Programming, Budgeting and Execution System (IPPBES). The NGA accounting process will be modified where necessary to allow for the easy identification and utilization of cloud cost resources by program.

The IC business model for IC ITE is based on a provider-consumer relationship in which the budgets to pay for IC ITE services reside with the consumers. IC ITE service providers are to be funded through customer payments, either by an approved cost allocation methodology for enterprise services, by a cost recovery solution (i.e., unit pricing) for commodity services, or through direct payments onto a multi-agency contract (e.g., C2S).

The CIO-T requirements management process ensures requirements are catalogued, reviewed and prioritized in coordination with the community. These requirements and costs will be validated and when approved, prioritized for resourcing as part of the yearly IPPBES workflow.

3.0 CLOUD FINANCIAL PLANNING

This section identifies the financial components of a cloud-based program and the methods/tools available for cost estimation. Prior to initiating cloud migration or a cloud-native program development good financial planning ensures a proper allocation of resources.

The first step in planning for cloud costs is to determine current IT related costs. Programs are to use the total infrastructure and application support funding for the procurement of cloud services and any additional funds anticipated for cloud migration. Relevant IT expenditures include:

- ***Infrastructure Service Provider (ISP) costs*** - ISPs provide compute, storage, network communications, and backup services. In the NGA data center environment, ISP expenditures were typically paid to NCE AIT and/or NCW IT/IS contractors. In the cloud, the CSP provides these services, which will be consumed and costed via variable usage fees.

- **Application Service Provider (ASP) costs** - ASP costs are associated with software development and upgrades. These services are typically provided by a development contractor other than the ISP. In the cloud, the ASP development and test environments should be provided by the CSP as part of its services.
- **Software Licensing costs** - The NGA Software Asset Management (SAM) Office is the resource for determining associated licensing costs. Approved software for the cloud can be found in the [SWAP List](#) or via discussion with the [SAM Office](#).
- **Miscellaneous costs** - Miscellaneous costs are associated with software development, operations and maintenance that have not been planned. Mission-driven changes in capability, capacity, and footprint that were unplanned are contributors to miscellaneous costs.

In the cloud, it can be expected that much of the ISP costs are to be apportioned directly into CSP costs. The majority of the ASP services costs remain with the ASP, with the exception of cloud-associated infrastructure costs which are to be transferred from the ASP to the CSP. ASPs are to provision their DevOps environments from CSP commodity resources.

CSP service costs are accrued on an on-demand, pay-for-use basis. Cloud services are to be provisioned on an as-needed basis, based upon developer and mission user requirements. This cost model is quite different from the native data center environment where costs are amortized over time regardless of utilization. Identification of actual DevOps utilization patterns and performance over time can provide the inputs for prudent planning and optimization of IT resources in a cloud environment.

NGA enterprise services (i.e., base services) are provided at no charge and are not included in the cost calculator. Likewise, GEOINT customers consume these services at no charge.

3.1 GUIDANCE

Guidance: Program Managers shall use the Cost Calculator to estimate monthly CSP costs.

Guidance: Cost responsibility for GEOINT consumer usage should be considered in infrastructure sizing and as part of the cost calculator inputs.

The PM is to use the CSPs current [Cost Calculator](#) which can be found on the [NGA Cloud Information Portal](#). The Cost Calculator spreadsheet contains content examples on how the form should be filled out. Please note that each domain and cloud have their own unique cost calculators, complete with their own unique service pricing model. It is imperative that the proper cost calculator be used when estimating costs for each cloud domain.

The cost calculator provides the most current service charge rates (i.e., periodic updates) for the CSPs available Amazon Web Services. Actual monthly charges will be based upon the metered usage of the delivered services and may vary from the estimates provided by the calculator. NGA's cloud support staff are available for assistance in applying data to the cost calculator.

The [Cloud Enterprise Services Catalog](#), located in the [ATLAS](#) contains the complete listing of NGA and CSP services.

Guidance: Cloud financial billings shall be managed for two environments, development with testing, and operations.

IC and DoD fiscal appropriations management necessitates that development with testing and the operational environments' respective costs estimates and expenditure billings be managed separately. This requires two separate cost calculator submissions. The type of funding (i.e., O&M, RDT&E, PDW) needs to be differentiated in the allocation of funds. Attention should be given to the unique requirements of each environment, specifically the number of users and hours of operations.

The [AF Form 9](#), which is equivalent to an Independent Government Cost Estimate (IGCE), needs to be processes for allocating CSP funds. The C2S contract requires certified funding for all C2S services for the entire period of performance. The AF Form 9 must pass PRISM/GEO-F validation and be routed in PRISM for approval in order to be added to the C2S contract. The [C2S CLIN Agreement](#) provides the domain and period of performance CLIN for C2S usage and training.

4.0 ENTERPRISE SERVICE AUTHORIZATION

This section describes how a given service fits in context of the enterprise. All common IC services are inherently enterprise services. All services follow essentially the same life cycle (e.g., ITIL), and are registered and catalogued. GEOINT services are NGA's valued enterprise services given their availability to multiple users, common data sharing, and multi-mission support.

Enterprise Systems Engineering & Integration² (ESE&I) is responsible for the integrating activity which develops and maintains the enterprise baseline. Additional detail on the ESE&I-managed NGA service lifecycle and systems engineering are provided in [Appendix B](#) and [Appendix C](#), respectively.

4.1 GUIDANCE

4.1.1 Enterprise Service Identification

Guidance: *Enterprise service candidates should be registered in the enterprise registry via Service Registry Board (SRB) action.*

Enterprise service candidates can be identified or nominated in a variety of ways, all of which result in a disposition of being either a service or enterprise service. The Service Registry Board (SRB) is NGA's control board for deciding if a nominated service becomes an enterprise service. The SRB represents the chief engineer and is responsible within ESE&I for maintaining the enterprise service registry and performs the adjudication of enterprise service nominations.

Guidance: *NGA services should meet the general service requirements (GSR) portion of the NGA requirements baseline.*

All NGA services must incorporate the following GSR information:

1. Alignment with NGA's architecture reference model (per JARM's TST notation)
2. Applies, as a minimum:
 - a. Common requirements from the enterprise (e.g., security, detached data, APIs).
 - b. Accredited service creation process and tools (i.e., Risk Evaluation Acceleration Management Process (REvAMP))

² ESE&I provides engineering integration, maintains NGA's enterprise baseline, and handles the technical aspects of NGA enterprise governance.

3. Identifies the service type as either program (internal) or enterprise (external) use
4. Administrative data (e.g., POC, service dates, functional description, funding source).

Guidance: *NGA services should be issued a registry ID and be published in the online and offline service registries.*

The NGA service registration process identifies new services which perform similar functionality in existing enterprise or program-centric services. Registrants may be notified to coordinate with other service developers to collapse or pursue joint service efforts. Adjudication of service collapse, joint, or other determinations and waiver requests are performed by the SRB. The SRB may also identify program services as enterprise candidates. The SRB examines new service registrations for overall value:

- User need or demand (e.g., priority)
- Overlapping service functionality (i.e., TST)
- Aligned with enterprise strategy (e.g., multi-use)

The SRB will confirm the initial registry ID or initiate the adjudication action. Viable service candidates to be “cloud ready” must incorporate the necessary GSR criteria and service information. Redundant services may be collapsed into, or replaced by, an enterprise service.

All new efforts are to be cloud-based services and waivers will be required for other solutions. GSR waivers and/or time constraints on temporary services can be levied by the SRB. Services may be encapsulated in an “accredited container” which supports the GSR via a waiver.

Services identified via redundancy analysis should have all redundant candidates evaluated against one-another and against the possibility of a ‘new start’ service. Such AoAs are to consider “time-to-capability” and mission value.

4.1.2 Funding and Sustainment

Guidance: *Programs shall be responsible for services support and funding, while ESE&I should maintain the mapping of mission-to-service-to-program for funding viability and planning.*

Services are assigned to a mission activity via a program’s capability alignment. Services are aligned as either enterprise or private services, and mapped to funding as billing line-items associated with their programs, as follows:

1. Development and test
2. Operations

Services are mapped to the JARM layer and one or more mission capabilities for requirements traceability in POM and FYDP activity planning, as shown in Figure 4-1.

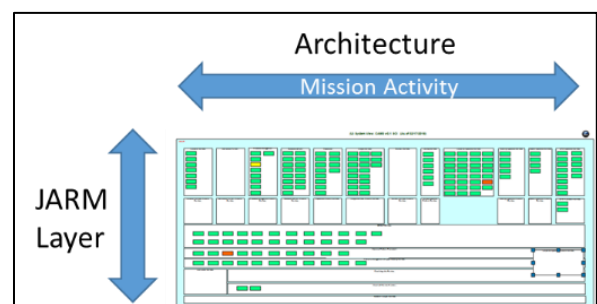


Figure 4-1 - Service Alignment

The development and “in-service” capability change dates (i.e., ATO, IOC, FOC) are used by ESE&I to trace the service’s capability evolution over time. The results are used planning purposes for sustaining or adjusting program funding.

4.1.3 Assessment and Authorization

Guidance: NGA services will perform the Assessment & Authorization (A&A) process in accordance with ICD 503.

Guidance: NGA services will use common standards and reciprocity in accordance with ICD 503.

Guidance: NGA services will weigh the application of protective security measures against delivering mission capabilities at acceptable risk levels.

The Assessment & Authorization (A&A) process for IC systems and services is managed under the IC policy established in [ICD 503, IC IT Systems Security Risk Management, Certification and Accreditation](#). In determining the level of acceptable operational risk at a particular security level, appropriate weight to mission and security requirements, information sharing, collaboration, and other relevant policy requirements must be evaluated. NGA is applying the REvAMP approach to its A&A security risk management.

Statement of Work (SOW) requirements are to include those for service Assessment & Authorization. Security requirements are to be tagged as security related in the GSR database, as well as those tagged as security-related in the SOW. Services may inherit the certification of services (i.e., C2S-accredited ISP services). As such, the framework on which the software runs, or services the software calls, being certified and accredited by the respective service provider can be applied through inheritance. This is particularly useful when COTS or GOTS function can be incorporated as a service component, which is already certified.

Figure 4-2 illustrates the scope of inheritance of requirements from the service's multiple GSR functional layers. The overall service A&A requirements are a combination of standards, unique capability features, security, and performance requirements for the service.

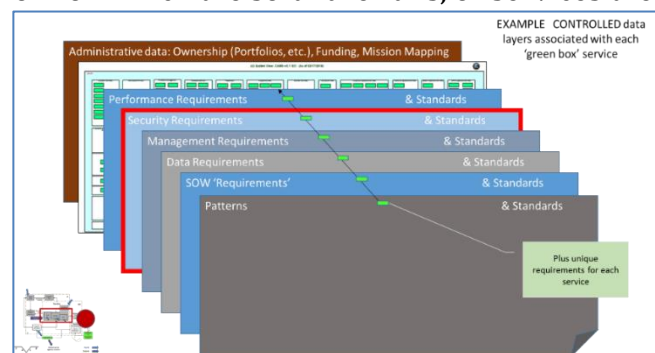


Figure 4-2 - Inherited Security Requirements

4.1.4 Lifecycle and Evolution

Guidance: Program Managers should register services as early in the lifecycle as practical.

Services should reflect the forecasted “in-service” and capability dates for the expected service lifecycle. The early identification of service capabilities helps the SRB with new start adjudication (e.g., service redundancy) as well as FYDP pre-coordination and funding.

Guidance: Program Managers should consult with the SRB and/or review the registry for current or scheduled services with similar functionality.

Services should consider development collaboration options (e.g., incorporating lessons-learned, or code re-use) during the service's concept and definition phases.

Guidance: Services should be registered prior to operations deployment.

Enterprise services operations may require additional IC ITE testing (e.g., common services) and community notification. The switchover from legacy infrastructures may require Community coordination prior to deployment (e.g., extended mission users).

Guidance: *Services shall provide configuration management during operational deployment.*

Operational users may be running on older versions of the deployed services or subject to networking constraints (e.g., cross-domain employment, extended users). Therefore, services are to:

- Ensure backward compatibility with prior APIs
- Support at least one version prior to the most current
- Notify users of significant functional changes and planned outages
- Provide reversible transitions between significant functional changes to minimize operational disruption. Note that load-balancing to transition between versions is the recommended approach.

4.1.5 Retirement

Guidance: *Services shall provide sustained, effective mission value to their users during operational deployment or be assessed for retirement.*

Retirement decisions are to be based on the mission value, technology obsolesce, and operational sustainment. Services contribute to mission value via their level of enterprise support to missions. Technological obsolescence is described as an 'S-curve' (see Figure 4-3) where continued investment becomes marginally less useful. This is directly applicable to infrastructure investments (e.g., data centers).

For the NGA cloud enterprise, continued investment in services to support a mission is driven by effectiveness to the mission and minimizing infrastructure investments. At some point in time, services may not be improved but may only be sustained operationally based on preserving mission value. Obsolete services will be discontinued in favor of services farther down on their S-curve with increased capability value to the mission. Mission stakeholder and user notifications are to be part of the retirement process when a service is being assessed for retirement.

ESE&I is responsible for documenting service capabilities and enterprise value for assessing retirement decisions.

5.0 CLOUD ENVIRONMENT PROVISIONING

This section identifies the necessary actions for the provisioning of an activity's Virtual Private Cloud (VPC). VPCs serve as the cloud's computational environment for the activity. Provisioning refers to preparing the VPC for use. Provisioning is complete once the user has been authorized and the VPC configured. Responsibilities for VPC preparation and provisioning are shared between:

- Cloud Service Provider (CSP) (i.e., C2S)
- Stakeholders (e.g., Program Manager)

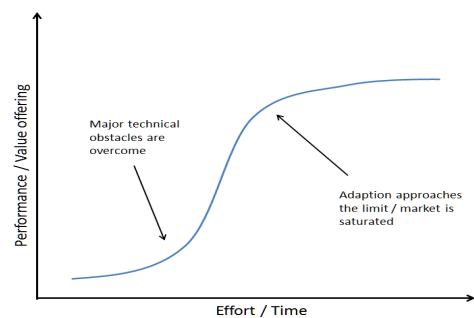


Figure 4-3 - Technology Obsolescence (S-curve)

- Cloud PMO

5.1 GUIDANCE

5.1.1 Preparedness - Launch (Phase 1)

***Guidance:** Program Managers and developers shall prepare the requisite architecture artifacts to ensure that the service implementation and software components are correctly described and configured for cloud implementation.*

The activity's provisioning process begins with the "Launch" (Phase 1). Phase 1 actions are largely the responsibility of the PM to prepare the requisite set of documentation for the Architecture Technical Exchange Meeting (TEM) with the Cloud team. The required Architecture TEM documents are as follows:

- **Cloud Adoption Template (CAT)** – provides standard program management and technical data, including points of contact, technical attributes, proposed transition schedule, funding details, and required services. Much of this data will be readily available from the PM, FM representative, and security lead.
- **Concept of Operations (CONOPS)** – provides a general system description and purpose of the activity, capabilities and requirements satisfied, information (data) types and roles and responsibilities required to develop and implement the activity.
- **System Security Plan (SSP)** – identifies the security attributes to include users, information types, and environment specifics to include ports and protocols, and security protection policies. This artifact is required for ICD 503 Assessment and Authorization.

The cloud implementation process starts with the submission and review of the Cloud Adoption Template (CAT) (i.e., [word, e-form](#)) and other artifacts. The CAT organizes the activity's preparation for the architectural meeting with the Cloud team. The completed CAT, [CONOPS](#), [SSP](#), and [AF Form 9](#) are to be forwarded to the [Cloud-Info Distro Group](#) for review and scheduling the architecture TEM.³

Templates for all documents are located on the [NGA Cloud Information Portal](#). Assistance can be requested of the [Cloud team](#) at [#NGA Cloud-Info](#).

All systems must adhere to the ICD 503 A&A process. The A&A registration meeting is to record the security requirements, key personnel and roles in XACTA. A Security Plan Identification Number (SPID) will be assigned. Registration meetings require a completed registration form (i.e., [Service Oriented Architecture Registration Template](#)), CONOPS and network diagram. PMs must coordinate with their assigned Information System Security Officer (ISSO) for scheduling the meeting. The A&A meeting may be held in concert with the Architecture TEM. Access the [XACTA](#) webpage for general A&A information (e.g., templates, workflows) and linkage to the [XACTA tool](#).

Phase 1 implementation requires coordination with the [Data Migration Group](#) to identify the data types, sources, and storage interfacing. If development uses the PaaS environment or GEOINT Services' tools, coordinate with the [GEOINT Services Group](#).

While not all documents are required to be complete prior to the conducting of the Architecture

³ [Phase 1 Checklist](#)

TEM, the following documents are required for completion of the provisioning process:

- **E-mail Distribution List** – provides the C2S commercial provider with the required contact list for the activity. Each activity must have a Development/Test, and Operations e-mail distribution list to coincide with their separate C2S accounts. Standard naming conventions for Distribution Lists are provided on the NGA Cloud Information Portal.
- **Roles Spreadsheet** – provides the activity's authorized cloud users and their respective privileges (e.g., read/write, read-only).
- **Cost Calculator** – provides monthly cost estimates for C2S services. Separate and distinct Cost Calculators should be developed for both Development/Test and Operations environments. The Calculator output provides the basis for the funding of cloud services.
- **AF Form 9** – provides the specifics regarding certified funding for C2S services. The AF Form 9 demonstrates commitment to operations and support costs and POM planning for full lifecycle costs. AF Form 9s must pass PRISM/GEO-F validation and approval.

Of these documents, the CAT, CONOPS and SSP are required for the Architecture TEM with the Cloud team. These artifacts provide information essential to the efficient sizing, CSP implementation and operation of the activity in the cloud.

5.1.2 Readiness - Architecture and Security Onboarding TEMs (Phase 2)

Guidance: The Cloud Adoption Template (CAT) and the architecture artifacts shall be reviewed at the Architecture TEM to ensure that the service implementation and software components are configured and correctly described for cloud implementation.

The Architecture TEM assist in building out the cloud architecture by a technical appraisal of the CONOPS, SSP and software requirements. The Architecture TEM is facilitated by Cloud team representatives to ensure the cloud implementation process is both understood and expedited. It is essential that the PM and technical lead be in attendance. The TEM objective is to provide SME guidance for achieving the "To-Be" cloud architecture, applying the appropriate services from the [Cloud Enterprise Services Catalog](#) and finalizing CSP cost estimates. The Architecture TEM coordination is initiated by sending an email to the [Cloud-Info distro group](#). The activity is assigned an Architecture TEM lead and a Delegated Authorizing Official (DAO) as their POCs for handling cloud questions. AWS Technical Account Manager (TAM) representatives are available for any Amazon related questions or concerns.⁴

Upon TEM completion, documentation is to be updated for the agreed upon cloud architecture, along with the corresponding cost calculation for review and approval by the [Engineering Review Board](#) (ERB). Assessments by two independent reviewers should occur within 48 business hours. If modifications or waivers are requested, a follow-up submission to the ERB may be required.

If the A&A meeting was not held in concert with the Architecture TEM, the [IC ITE Security A&A team](#) and [Risk Management](#) need to be contacted to determine the appropriate security controls via a Security Onboarding TEM (SOT).

⁴ [Phase 2 Checklist](#)

The SOT provides the information and tools necessary to efficiently and successfully navigate through the Risk Management Framework (RMF) process.⁵ This includes:

- Overview of the RMF process and general A&A requirements
- Specific requirements for the Categorization and Selection TEM (CAST)

The Categorization and Selection TEM (CAST) meeting reviews the data types being processed, resulting in the determination of a security categorization value of the Confidentiality, Integrity and Availability for the program; as well as, the applicable security overlay (e.g., H-M-M, INT-A) controls. The CAST will address conformance to the enterprise audit requirement, Ports Protocols and Services (PPS), boundary diagram, and the Privacy Threshold Analysis (PTA) form. Programs with an existing SPID will receive a new 6000 series number.

Upon completion of the Architecture and the CAST TEMs, including any provisioning liens, the activity's C2S account is established and the Authorization to Proceed (ATP) granted. A process flow diagram for getting an ATP is shown in Figure 5-1. Reference the [PM's Guide to Cloud Migration](#) for the detailed procedures.

Guidance: *The completed Cost Calculator shall be attached to the AF Form 9 as evidence for the required funding to establish the CSP (e.g., AWS/C2S) account(s).*

Upon completion of required documentation, the Cloud team establishes the AWS/C2S account(s) for the activity (i.e., Dev/Test, Operations). The AF Form 9 must be completed and approved by all required stakeholders prior to account creation. Account requests are made by the Cloud team via the [Joint Store Front \(JSF\)](#) and are “approved” by:

- Security (CIO-T/TS) for validity of requested accounts
- Business (CIO-T/TIIR) for confirmation of authorized funding

All approvals are required for a C2S account authorization to proceed. Denials result in follow-up actions by the PM.

Upon approval, the Cloud team prepares the necessary cloud instances for the activity, configured per the cloud “To-Be” guidelines. A C2S kick-off meeting is held to discuss VPC maintenance and general C2S operations support, if required. The VPC is now considered in “fully provisioned” status and available for activity configuration and development by the PM, as shown in Figure 5-1 **Error! Reference source not found.**, with the granting of an ATP.

⁵ [Information Assurance Requirements Catalog \(IARC\)](#)

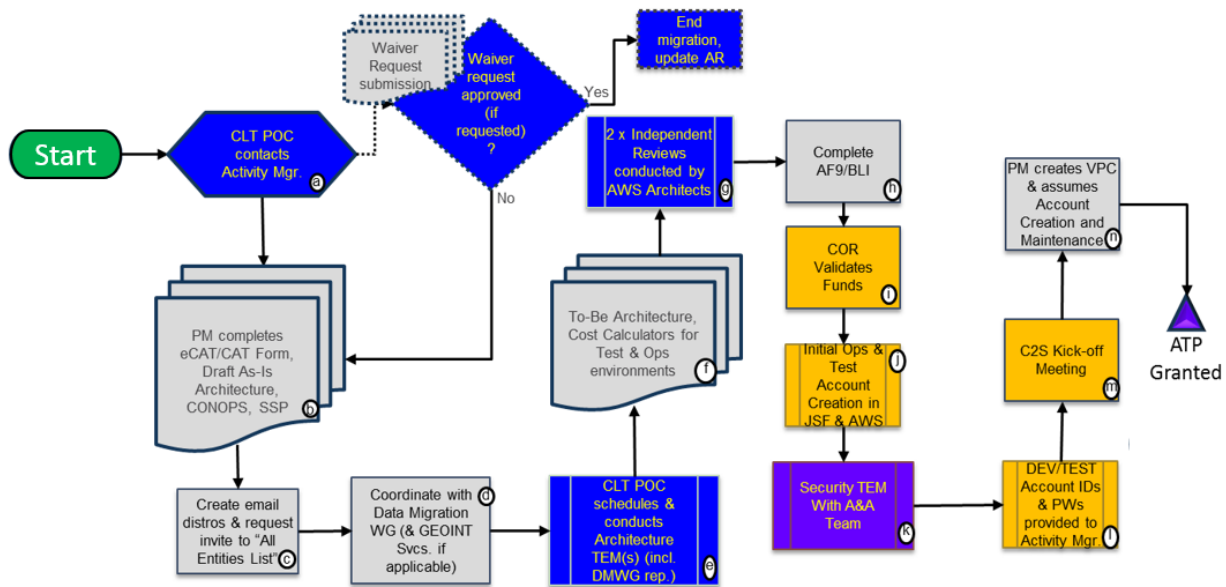


Figure 5-1 - Provisioning Process Flow

An ATP memorandum is prepared to acknowledge the security categorization of the program and formally authorize development for implementing operations in the cloud. Note that no expenses are incurred until the account is used.⁶

5.1.3 Implementation (Phase 3)

Guidance: Program Managers shall prepare the requisite documentation for IATT and ATO approval.

Provisioning establishes the cloud environment for the activity's service development and/or operations. The VPCs and associated data feeds are maintained by the activity's development staff via the [C2S Access Portal \(CAP\)](#). The [Consolidated Service Portal \(CSP\)](#) provides the means to generate a request of data transfer and obtain additional C2S services (e.g., cross domain). If [GEOAxIS](#) is to be used for authentication, contact the [GEOAxIS team](#) to coordinate its integration.

NGA periodically publishes a set of security compliant Amazon Machine Images (AMIs) to be used by NGA programs for their Elastic Compute Cloud (EC2) instances. The AMI provides the necessary AWS configuration to launch an EC2 instance with NGA base security, DISA Security Technical Implementation Guides (STIGs), and hardened application configurations applied. It is the responsibility of the program to complete all A&A activities.

AMI versions are maintained for two iterations. Any AMIs older than two iterations will not be supported. Additionally, only base AMIs will be supported. Any add/modify/change of the AMI by the program will not be supported by C2S operations.⁷

⁶ [Phase 3 Checklist](#)

⁷ [NGA AMI Release Notes](#)

5.1.3.1 IATT Process

The required documentation for IATT approval follows:

- CONOPS update with a review and incorporation of any [Risk Management](#) feedback.
- SSP update with Risk Management and TACC feedback.
- System Accreditation Plan (SAP) via TACC.

Figure 5-2 shows the document updates required prior to IATT approval (i.e., Draft CONOPS (>80%). Also, the Security Controls Traceability Matrix (SCTM), Information System Control Matrix (ISCM), Access Control Model Framework (ACMF) audit, the Software Assurance Process (SWAP) artifacts) and a “clean” vulnerability and compliance scan, are the requisite body of evidence (BOE) for the IATT.

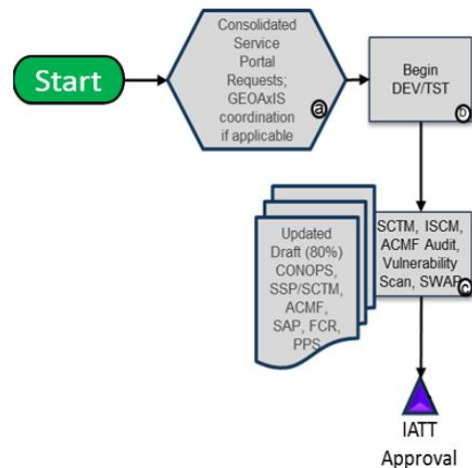


Figure 5-2 - IATT Process Flow

Request ACAS vulnerability and compliance scans from the [TST Request Portal](#) and to set up a scan. A “clean” scan implies there are no critical or high security findings. All critical or high findings are to be mitigated, with the IATT approval or waiver by the DAOR. Reference the [PM’s Guide to Cloud Migration](#) for the detailed procedures.

An IATT allows the program to: 1) configure security groups and NACLs for connections outside the VPC, 2) integrate with the ESS services, and 3) begin integration and functional testing.

5.1.3.2 ATO Process

The required documentation for ATO approval follows:

- SCTM via Security
- Audit strategy document per program deployment architecture
- Configuration Management Plan (CMP)
- Security Control Assessor (SCA) Assessment
- Information System Authorization via Security
- ATO Decision Brief

The DAO/DAOR will assess the activity's documentation package for the ATO decision. Reference the [PM’s Guide to Cloud Migration](#) for the detailed ATO procedures.

6.0 CLOUD BILLING AND MONITORING

This section describes the cloud billing cycle to include sequencing and timing of events, stakeholder roles and responsibilities, and financial dependencies. Responsible parties for a program’s cloud funding, billing, and financial monitoring follow:

- Program Manager (PM)
- Financial Management (FM)
- Commercial Service Provider (CSP)

- Cloud Team

6.1 GUIDANCE

Guidance: Program Managers and their Financial Management (FM) representatives shall be responsible for the program's CSP (e.g., AWS/C2S) account(s) cost management.

Guidance: User role assignments and invoice processing shall be managed by NGA's Cloud Program Management Office (PMO).

Guidance: Financial Management (FM) shall authorize the release funds to the CSP upon invoice certification.

AWS/C2S costs are available at all times to authorized users via the Amazon provided [Cloud Access Portal \(CAP\)](#). CAP uses role-based authorization via [AccessIT!](#) to grant access to the accounts. Role assignment is managed by NGA. Each program should have at least a BILLINGMGR role to access the billing data each month. The [CAP User Guide](#) provides detailed information on the CAP and troubleshooting suggestions.

C2S accounts are to be registered with CAP and assigned a unique mission name in order to support 'Federated Access' capabilities. CAP provides administrative features that allow NGA to manage its C2S accounts, user roles, and account settings independently. Account set-up activities are performed by NGA's Cloud PMO based on requests from the [Joint StoreFront \(JSF\)](#).

Note that with consolidated billing, only the master account can create and manage budgets. Individual member accounts cannot create or manage budgets; member accounts have read-only access to budgets in accordance with the [Controlled Access](#) policy.

The [Billing and Cost Management console and dashboard](#) interface provides authorized users with the AWS/C2S cost data for the program. The console opens to the dashboard, where current month-to-date usage graphs are displayed. The dashboard function allows for easy access to last month's expenses, current month-to-date costs, as well as budget forecasts based on historical monthly costs (i.e., Cost Explorer).

AWS bills accounts only for usage. To help manage expenditures, AWS provides several easy-to-use features that help account management, including [alerts and notifications](#), cost allocation tags for detailed accounting, and cost optimizing (i.e., [Trusted Advisor](#)).

For a cost management example, the CloudWatch billing alarm notifies the manager when costs exceed the set budget threshold, as shown in Figure 6-1. Figure

Trusted Advisor inspects the program's operating environment and recommends opportunities to save money and improve system performance via best practices. Also, as a cost saving best practice, consult the Design White Paper - [Turn Off Compute When Not Needed](#). For more information

Create Alarm

Billing Alarm

You can create a billing alarm to receive e-mail alerts when your AWS charges exceed a threshold you choose. Simply:

1. Enter a spending threshold
2. Provide an email address
3. Check your inbox for a confirmation email and click the link provided

When my total AWS charges for the month exceed: \$ USD

send a notification to:

Reminder: for each address you add, you will receive an email from AWS with the subject "AWS Notification - Subscription Confirmation". Click the link provided in the message to confirm that AWS may deliver alerts to that address.

showing simple options | [show advanced](#)

Alarm Preview

This alarm will trigger when the blue line goes above the red line.

EstimatedCharges > 1000

1,200
1,000
800
600
400
200
0

0:10 0:12 0:14 0:16

More resources

- [AWS Billing console](#)
- [Getting started with billing alarms](#)
- [More help with billing alarms](#)
- [AWS Billing FAQs](#)

Figure 6-1 - Billing Alert Threshold Worksheet

about budgets, see [Managing Your Costs with Budgets](#).

Costs for CSP services are accrued, invoiced, and paid on a monthly basis, as illustrated in Figure 6-2. The invoices are generated and validated for accuracy, completeness, and available funding by NGA's Cloud PMO. Once the invoice is deemed accurate, the Cloud PMO certifies the invoice and delivers it to FM for payment. Funds are debited from a program's available funding.

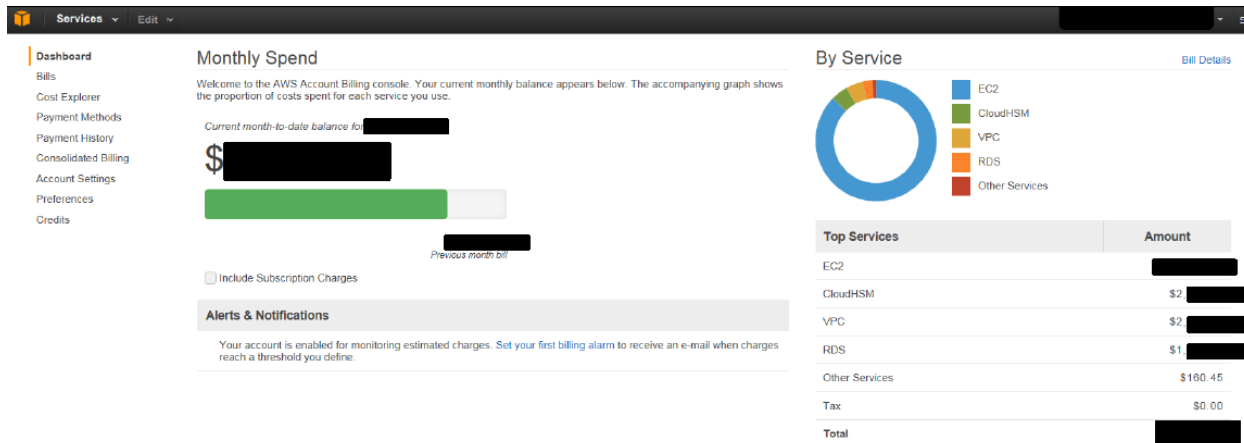


Figure 6-2 - C2S Monthly Bill Format

7.0 ACTIVITY OPTIMIZATION

This section provides guidance for the PM, developers and service operations staff regarding a program's cloud activity performance and cost optimization. Key cloud operating principles, as well as, performance metrics and utilization tools will be discussed.

The AWS/C2S provides utilities that monitor performance and service utilization data of its resources. Periodic review and analysis of this information enables users to make informed decisions on optimizing cloud resources and reducing program costs.

Cloud architectures provide on-demand, elastic provisioning of resources. Properly architected and sized cloud resources can readily meet demand, even under peak operational loading conditions. Appropriate sizing of a CSP service often results in significant cost savings. Also, elastic provisioning is especially advantageous in development/test environments where use is primarily during business hours. If an instance is turned off, it's not accruing running charges; however, a fine point is that Elastic Block Storage (EBS) attached to that stopped instance does still accrue charges. Shutting down unused instances (e.g., EC2s) results in noteworthy savings. Consult the Design White Paper - Elastic Block Store for best practice recommendations on reducing EC2 operating costs.

7.1 GUIDANCE

Guidance: Program Managers and developers shall not replicate existing legacy infrastructure requirements or design configurations in the cloud.

Cloud environments operate differently and more efficiently than legacy data center environments. Replicating non-elastic legacy infrastructures that are designed for "peak use" or "DR back-

up” conditions, will result in supporting unnecessary infrastructure, over scaling and increased cost. Cloud resources are scalable and programs pay for only what is in use.

Consult Annex F: Cloud Design Guidance for a description of software architectures, reference models and architectures for cloud applications.

Guidance: *Program Managers and developers shall use performance metrics, optimization tools and analysis recommendations to increase the program’s cloud operating efficiency and reduce costs.*

The CSP and Cloud PMO provide a set of management tools which allow users the visibility and ability to modify its operating environment. From the [AWS Management Console](#), authorized users (e.g., PM, BILLINGMGR) can control and monitor cloud services, run both standard and custom reports, establish alerts/thresholds, and receive and view AWS automated alerts identifying opportunities for system optimization.

Trusted Advisor

Trusted Advisor automatically inspects the cloud operating environment and provides best practices recommendations for four categories: cost optimization, performance, security, and fault tolerance, as shown in Figure 7-1. The tool provides ~29 separate check functions and offers adjustments per industry best practices. [Trusted Advisor](#) is available to users directly from the AWS Management Console. Consult the C2S Trusted Advisor page for additional information.

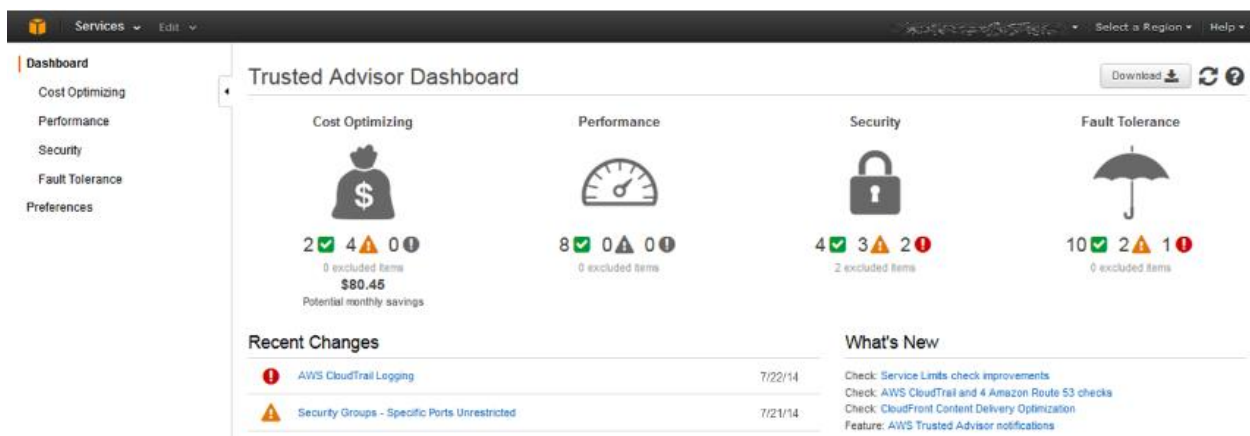


Figure 7-1 - Trusted Advisor Dashboard

Trusted Advisor inspects each program’s operating environment and recommends opportunities to save money and improve system performance via best practices. Trusted Advisor reporting identifies periods of inactivity, whereby instances should be turned-off. Consult the Design White Paper - Turn Off Compute When Not Needed.

CloudWatch

CloudWatch provides monitoring of AWS cloud resources and programs’ applications. Developers and SAs can use it to collect and track metrics and react immediately to keep their applications optimized. The tool monitors AWS resources (e.g., EC2) and can use custom metrics for applications and services. For more details about custom metrics, consult the [Amazon CloudWatch Developer Guide](#).

Amazon CloudWatch provides system-wide visibility into resource utilization and application performance. It provides metrics collection and reporting, and generates notifications, automated alarms and alerts. [CloudWatch](#) is directly available from the AWS Management Console.

CloudTrail

CloudTrail enables security analysis, resource change tracking, and compliance auditing. CloudTrail provides a history of the program account's API calls via logs, which permit a SA to review usage patterns. The recorded information includes the identity of the API caller, time of the API call, source IP address of the API caller, request parameters, and the response elements returned by the AWS service. By default, CloudTrail encrypts all log files delivered to the specified S3 bucket using server-side encryption (SSE).

This data can be used to track mission support and tune program cost. [CloudTrail](#) is also directly available from the AWS Management Console.

Cloud Utilization and Optimization (CUO) Reports

The Cloud Utilization and Optimization (CUO) team provides an account specific report to PMs offers engineering analysis and management oversight into a program's AWS provisioned cloud environment. The report provides a visualization and consolidation of information collected from Amazon Tools: CloudWatch, Trusted Advisor, CloudCheckr, and SEIN Cloud Financials in order to assist an activity in optimization efforts. The report provides data from AWS resources: EC2, EBS, RDS, and S3 for a given account and provides detailed recommendations for resource changes, tailoring, and improvements. For more details on CUO reports, consult the Design White Paper - Cloud Utilization and Optimization or email: [#NGA Cloud O&U Team](#) for assistance.

8.0 CLOUD SOFTWARE LIFECYCLE AND LICENSING

This section provides guidance for the PM, developers and service operations staff regarding software assets to include approved software versions, licensing, deployment and availability in the cloud environment. This guide identifies the process and procedures of software employment in the areas of: planning, operating and retiring software. Software assets include all Commercial-Off-The-Shelf (COTS), Government-Off-The-Shelf (GOTS) and Free and Open Source Software (FOSS) used within the cloud environment. Some of the basic assumptions for software use in all stages of employment are:

- Software license acquisition and maintenance costs are in the current and out year budgets.
- Software acquisitions will be coordinated with the SAM Office via the [ARTS](#) tool.
- All software will be maintained and deployed in compliance with SWAP and software licensing agreements.

The key operating tenets for software assets used within the cloud are:

- Software license agreements⁸ are legally binding contractual agreements that define conditions of use (e.g., COTS and Open Source license agreements).
- Software is provisioned for use per the contract's Terms & Conditions with the vendor.

⁸ See [Appendix D - Software License Definitions](#)

- Not all software license use rights are transferable to a cloud environment (e.g., Microsoft desktop applications such as MS Office products).
- Enterprise License Agreement (ELA) and Unlimited License Agreement (ULA) do not mean licenses are free or that they necessarily have truly unlimited use.
- Bring Your Own License (BYOL) is not a one-time cost; annual maintenance billing is required to maintain currency of software functionality and security patching.

8.1 GUIDANCE

8.1.1 Planning

Guidance: Program Managers and developers should conduct research on licensed software availability to support a program's operation in the cloud.

Guidance: Program Managers shall document all vetted software needs on the Cloud Adoption Template (CAT).

The program's software needs are to be identified in the [Cloud Adoption Template \(CAT\)](#), System Security Plan (SSP), and CONOPS in support of the Authentication & Accreditation (A&A) process. Update the cloud deployed software versions to those approved on the [Software Whitelist Assurance Process \(SWAP\) listing by Software ID \(SWID\) number](#). All software contained on latest SWAP list is approved for use on all NGA network domains.

The program's proposed software assets are to be listed in the CAT's Software Asset Management section, as shown in Table 8-1, including any GOTS and FOSS in use, except the specific Operating System (OS) (e.g., Windows, RHEL) as shown in Application Name since OS is normally part of the Amazon Machine Image (AMI) unless a custom OS is required. Add to the list any new software assets to be used in the cloud (e.g., Application Development Services (ADV), GEOINT Services, IC Marketplace). Software research support can be requested from the SAM Office.

It is recommended, as part of this planning phase, follow the following prioritization in determining your software requirements for deploying into the cloud; in particular, if you are migrating from a NGA datacenter:

1. [Cloud Enterprise Base Services Catalog](#) offerings – many are mandatory services (e.g., enterprise audit, GeoAxis).
2. [GEOINT Services](#) offerings.
3. [Application Development Services \(ADV\)](#) offerings.
4. Current C2S Services Offerings such as Database Services (e.g., MySQL, DynamoDB, Oracle (BYOL)) and Simple Queue Service.
5. [IC Marketplace](#) offerings for evaluating software alternatives (not for deployment into production without SAM Office coordination).
6. Current licenses (BYOL) used in the NGA datacenter if appropriate – dual operations license restrictions.
7. Refactor for FOSS use (rather than using COTS licensed software).

The software list baseline provided in Table 8-1 (Example) will be reviewed by the [Software Asset Management \(SAM\) Office](#) to validate the software assets planned for deployment in the cloud environment are compliant with software license agreements. If the test software is to be maintained in an enduring Dev/Ops environment, then licenses will need to be purchased unless the

Terms and Conditions (Ts&Cs) allow for no-cost test and evaluation licenses (not typical). The SAM Office can help determine how to “count” license requirements for all COTS licensed software in particular, processor-based licenses. If new licenses needed to be procured, the program will need to submit a software request via the [Acquisition Requirements Tool for Software \(ARTS\)](#). As the program progresses through IATT and A&A process, the software baseline will most likely change therefore to ensure license compliance a snapshot of this revised baseline should be coordinated with the SAM Office at least 60-days prior to ATO as a minimum and SSP updated as appropriate.

Table 8-1 - CAT Software Asset Form (Example)

Name	Version	Make	Type	Environment (Legacy/Test/Ops)	Software ID (SWID)	License Count
(Linux) Apache HTTP Server (httpd-x86_64)	2.2.15-30.el6_5	Red Hat, Inc.	FOSS	Test/Ops	12345	#
(Windows) Oracle Enterprise Database	12.0.2	Oracle Corp.	COTS	Test/Ops	56789	# (processor-based)

The program needs to verify the SWID via SWAP for each software asset that will be used in the Dev/Test or Ops environment. [Request for new software](#) or new versions of software will need to be processed through the on-line SWAP process. If there is no SWID for the software, enter “SWID Required” in the SWID Column. This table may be copied into the [System Security Plan \(SSP\)](#).

When completing the CAT (under Cloud Services), the program will need to request for software licenses (e.g., ESRI ArcGIS) requiring a MAC address the requirement for Elastic Network Interface (ENI) configuration for their Virtual Private Cloud (VPC) in order to deploy any MAC-based licensed software in the cloud.

If unsure about a software asset, discuss with the SAM Office what and how software can be provisioned into the cloud. The SAM Office has an engineering team to provide assistance with the assessment of new software, the sun-setting of software, and analyzing vendor provided data. The SAM Office will check with the vendor regarding any potential issue with cloud-based reuse such as any restrictions on desktop applications and dual operations use of the existing NGA procured software. Typically, there are 30- 60 days of dual operations use depending on the contract’s Ts&Cs with the vendor. If longer dual operations use is required, the SAM Office must notify the vendor and this may incur additional program costs.

A program’s software manager and/or developers should be aware of any software applications (in particular software bundles) that contain third-party applications and so annotated on CAT so the SAM Office can assess if migrating into the cloud invalidates the third-party license agreement.

If the SAM Office does not have a contract with a software vendor, then a Justification and Approval (J&A) statement must be prepared by the program and approved by NGA General Counsel for open competition in accordance with federal law. The SAM Office can assist the program with drafting the J&A supporting the procurement of the required software.

Guidance: *Program Managers shall transfer software licenses previously acquired by the program outside of the SAM process to the SAM Office.*

The software license transfer is initiated by e-mailing the Directorate [SAM Account Manager](#). The Account Manager will send a software transfer spreadsheet to be filled out. The program will need

to provide the SAM Office with proof of ownership (e.g., Purchase Order and Period of Performance of current licenses). The SAM Office's engineering team and SAM account managers are knowledgeable on software licensing models (e.g., floating, perpetual, enterprise) and their application for a variety of software applications and vendors.

8.1.2 Operations

Guidance: *Program Managers shall coordinate the use of conditional software licenses for test purposes.*

The program may use conditional (temporary) licenses from vendors during development for testing and integration prior to going operational in any cloud domain. Typically, these would be "term" licenses for a given timeframe. This requires a letter from the specific software vendor sent to the SAM Office stating that receipt of these licenses does not obligate the government for any future procurement action and these "term" licenses must have a specific termination date (90-120 days) that will stop operating after this date. Note that there may be a cost for use of these term licenses depending on the Ts&Cs. At the conclusion of testing and prior to going operational, the program will need to provide proof to the SAM Office that "term" licenses are removed from the Dev/Test environment.

Guidance: *Program Managers shall remove test and existing data center licenses once an activity moves into cloud operations.*

If running under a dual operations condition, deploying licenses into operations within the cloud requires test VPC and existing NGA data center instances be turned off (and licenses removed) at the end of this period to ensure license compliancy. The SAM Office requires proof that test VPC and data center instances and licenses are removed unless an enduring test VPC has been established and appropriate licenses procured. The dual operations timeframe must be coordinated with the SAM Office and contracts to negotiate any T&C changes for dual operations.

Guidance: *Billing for the use of software licenses shall be maintained by the CSP and the SAM Office.*

Each program (by VPC account) will receive a monthly bill from AWS via the NGA Cloud PMO for their utilization of cloud services, to include any software services requested through the IC Marketplace.

Annual software renewal billing is sent via an N-CERTS Tasker (e.g., *N-CERTS Tasker NGA-U-2017-####: FY17 SAM Maintenance Bill to NGA/###*) to the Directorate for their current software inventory, including any BYOL and term software licenses running in the cloud. For each PM and the Directorate, a SAM Account Manager is assigned to work the software renewal procurement activities during the annual billing process.

As part of this process, the SAM Office re-verifies the accuracy of the software inventory. However, this does not include software that an integration contractor may have purchased for a program unless they have transferred the licenses to the SAM Office. Note, the SAM Office has found that integration contractors have often failed to purchase the necessary one-year maintenance for the software as required by NGA policy. This forces NGA to purchase new software licenses since the original license was not maintained properly.

Guidance: *Program Managers should conduct periodic reviews of software usage.*

The PM should track and periodically review software usage to determine if the software is still required in the program's baseline. Particular attention should be given to data center license reductions where possible, BYOL (perpetual and subscription-based licenses) and temporary licensed software as the cloud instantiation reaches the ATO, IOC, and FOC milestones. As the baseline is changed, the program's software baseline should be updated in the SSP and provided for review to the SAM Office to ensure proper licensing and license compliance per deployment. Additionally, periodic review of software usage is important to ensure that the program is adequately licensed as demand may increase over time (e.g., additional EC2 instances stood up).

8.1.3 Retirement

Guidance: *Upon program retirement, software licenses shall be terminated or repurposed by the PM; and, all software binaries and keys shall be turned over to the SAM Office prior to hardware being turned off.*

The PM should notify the SAM Office no less than 120 days prior to initiating the Decommission and Disposal Review (DDR) process for a program and will ensure that licenses and/or license keys are provided to SAM Office prior to the activity's removal from the datacenter or cloud environment as appropriate. SAM Office coordination will ensure the proper adjudication of software license agreements at program retirement. Many software licenses have been re-purchased or maintenance continued due to lack of harvesting of software licenses during this process. This includes server OS licenses (e.g., RHEL, Windows Server) in the datacenter. In the DDR process, a decommissioning CRQ is to be submitted to the Infrastructure Requirements Approval Board (IRAB).

APPENDIX A - PROGRAM BUILD MILESTONES AND EVENTS

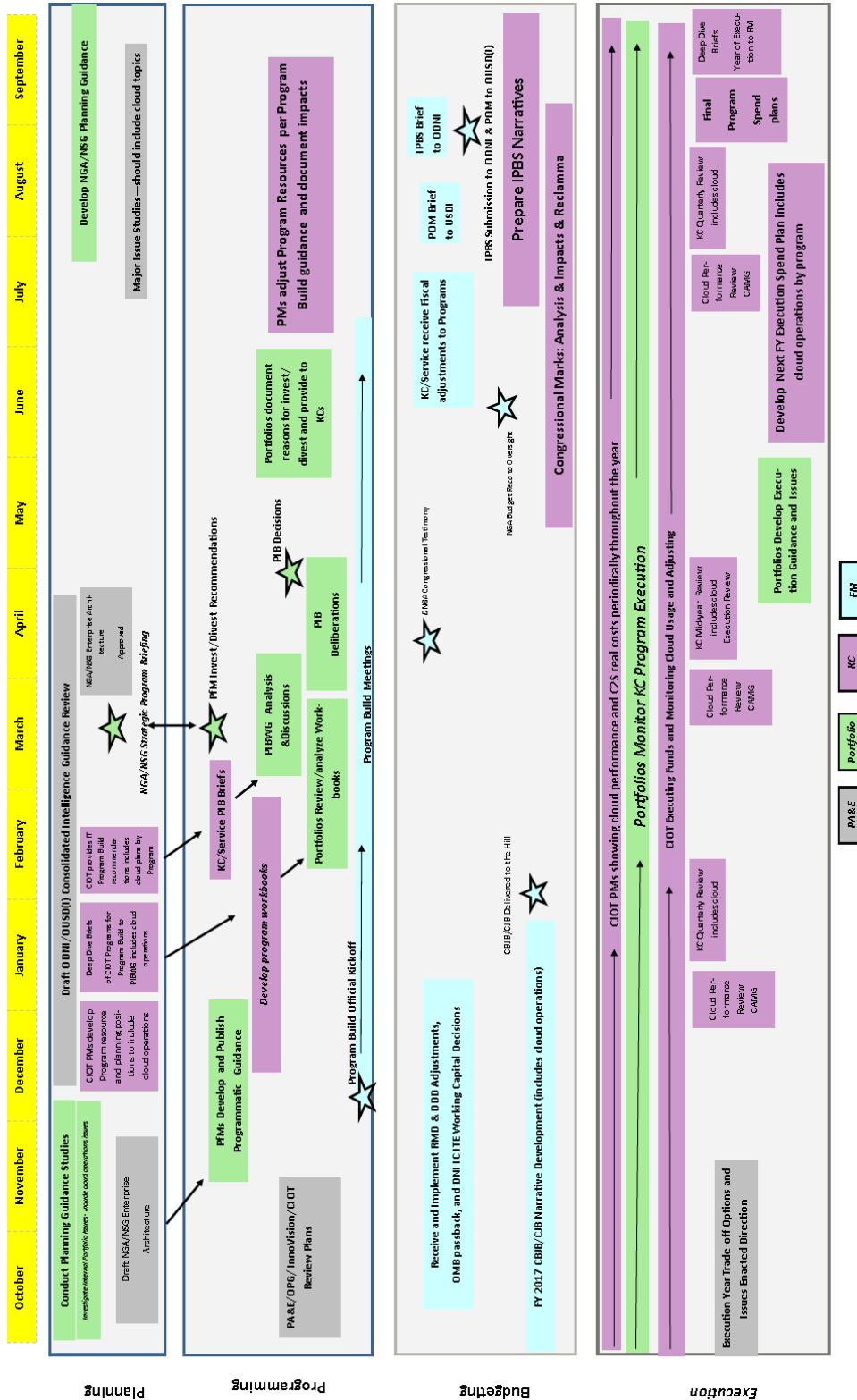


Figure A-1 - Program Build Schedule (Nominal)

APPENDIX B - SERVICE LIFECYCLE ARTIFACTS

B-1 Service Management Artifacts

The following examples depict where a service fits within the enterprise, and where relationships exist between services, missions, and programs. The figures are explained briefly for context and to provide an overview sketch of the service lifecycle from an enterprise management perspective.

The definition of these relationships, hierarchies and vocabulary are owned and controlled at the enterprise level. The data items and relationships themselves are owned and controlled by their respective owners (e.g., program) but managed and arbitrated by ESE&I to produce optimized results at the enterprise level.

Figure B-1 shows a cloud-based enterprise framework populated by services (green [and other] colored boxes). The grid is arranged horizontally by mission activities (e.g., TCPED). It is arranged vertically by IT layer; networks and compute/storage cloud hardware at the bottom, common services in the middle (horizontal white zones), domain level frameworks and mission software at the top. Other data relationships will be shown on-top-of, or using this framework.

Figure B-2 shows activity capabilities mapped across the framework via threads (e.g., yellow dots) for correlation of NGA service with the supported missions. A large number of such traces have overlap with common services. These traces relate services to mission capabilities. The figure depicts top level service requirements flowing through acquisition to development and into operations.

Graphics and Artifacts

Figures depicting services and relationships within the enterprise framework are graphical representations of data structures maintained by ESE&I.

Data structures are managed under enterprise level configuration control for decision-making based on the respective services, missions, and relationships.



Figure B-1 - Cloud Enterprise Framework Services

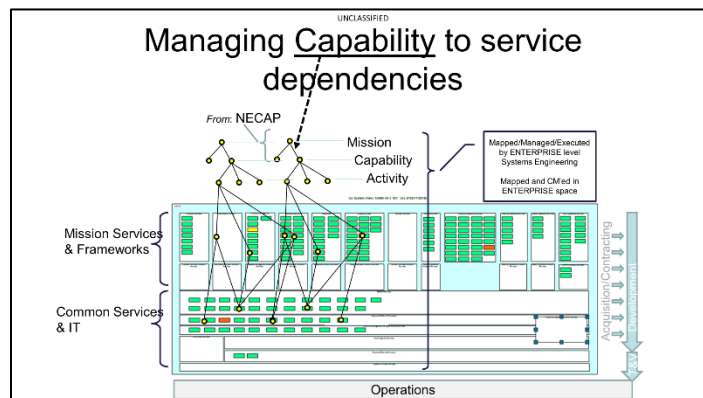


Figure B-2 - Mapped Mission to Capability Threads

Service requirements and the metadata associated with a particular service within the enterprise, is illustrated in Figure B-3. Requirements are inherited by a service when a service-dependent component is applied. Also, general requirements (i.e., standards) are assumed before any unique or specific function or capability. The service's metadata contains ownership, funding, and mission mapping data.

As services are added or upgraded within the enterprise framework, their change dates become part of the lifecycle-associated metadata.

Figure B-4 illustrates the time ordering of service upgrades reflected in an enterprise level schedule and associated capability roadmap. The capability evolution (i.e., Roadmap) is based on “value-based” improvements to the mission, as shown in Figure B-5. Roadmaps enable the visualization of investments to services over time. The service owners generate and manage individual schedules while ESE&I integrates the enterprise capability/value data.

Enterprise-level mission value over time is reflected in the value roadmap, also shown in Figure B-5. ESE&I's function is to help agency leadership maximize investments' value and NGA's contributions to the IC and DoD missions.

When a service's mission value has flat-lined or is declining, or the value of the capabilities it supports have flat-line or are declining, then it is time to transition to a newer technical capability and retirement of the legacy service. Figure B-6 illustrates the technical lifecycle phenomenon (i.e., value/time S-Curve).

The measurement of services and their supported mission contributions to overall enterprise value must be weighed against all other missions and services in light of available resources.

ESE&I will manage service-by-service data to allow the stakeholders to make capability trades and service lifecycle decisions. Service owners maintain their respective data.

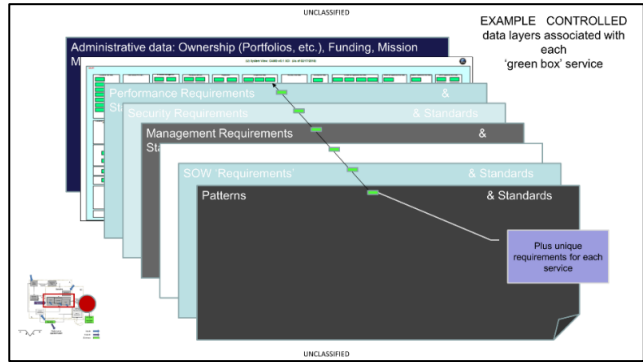


Figure B-3 - Service Requirements and Metadata

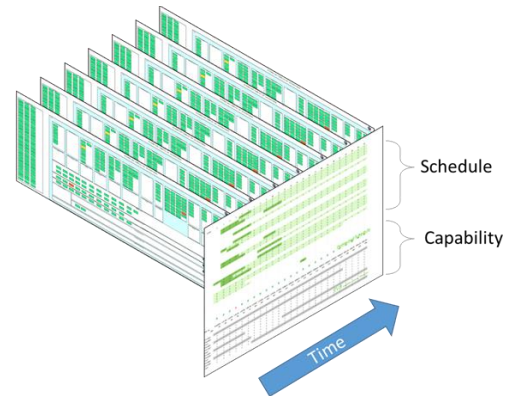


Figure B-4 - Capability Framework Over Time

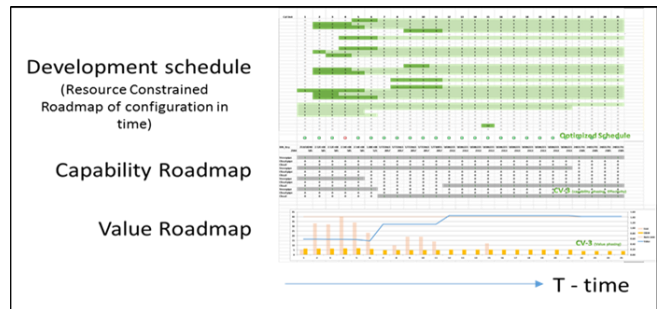


Figure B-3 - Enterprise Schedule, Roadmap, and Value Over Time

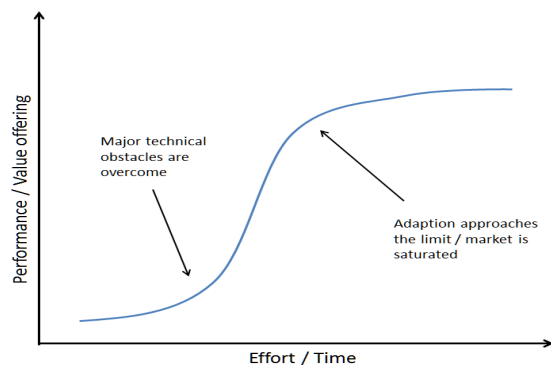


Figure B-4 - Tech Lifecycle S-curve

B-2 Service Registration and Cataloging

In order for a service to be visible to other users it must be cataloged and registered for discovery (e.g., TST service type) All service owners are responsible for keeping their catalog metadata current (e.g., capability upgrades, functionality, APIs, mission).

Mission alignment and mapping links service components with mission, and service elements (e.g., yellow dots/lines) is shown in Figure B-7. Capability mapping provides traceability between services (i.e., capability) and missions (i.e., value).

Service owners develop the service catalog data and associated metadata and ESE&I maintains the data for the enterprise. The coordination and maintenance of this information at the enterprise level is key to capability planning and sustainment.

If the service was nominated from a COTS or GOTS package or similar stand-alone software, it will be necessary to integrate, at which point the General Service Requirement (GSR) inheritance factors (i.e., standards) are applied, as shown in Figure B-8. This includes requirements for:

- Licensing
- Security
- Interoperability
- Encapsulation (e.g., API, containers)

These requirements can be thought of as basic ‘stem cell’ functions of a service component. Maintenance of this requirements process and the information allows ESE&I to characterize and track enterprise service’s capabilities.

B-3 Cloud Adoption “as-a-service” Philosophy

NGA is evolving the ASP/ ISP operating models towards an ecosystem with CSPs delivering enterprise-level services. As NGA begins to implement enterprise services, this invokes changes in the

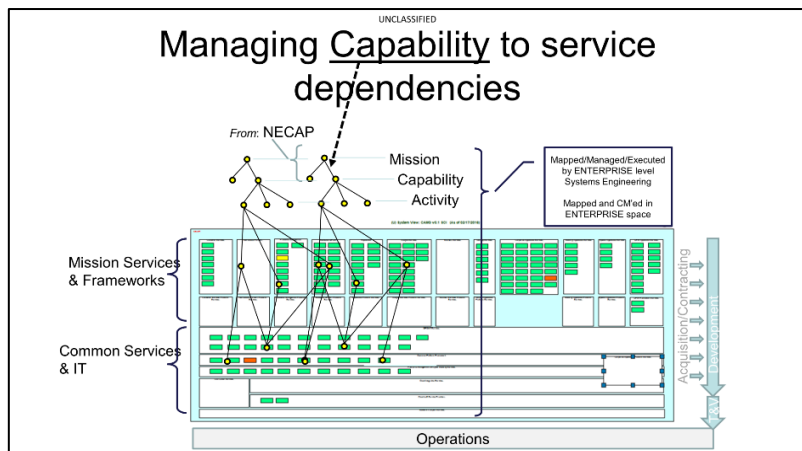


Figure B-5 - Inserting a Service into the Framework

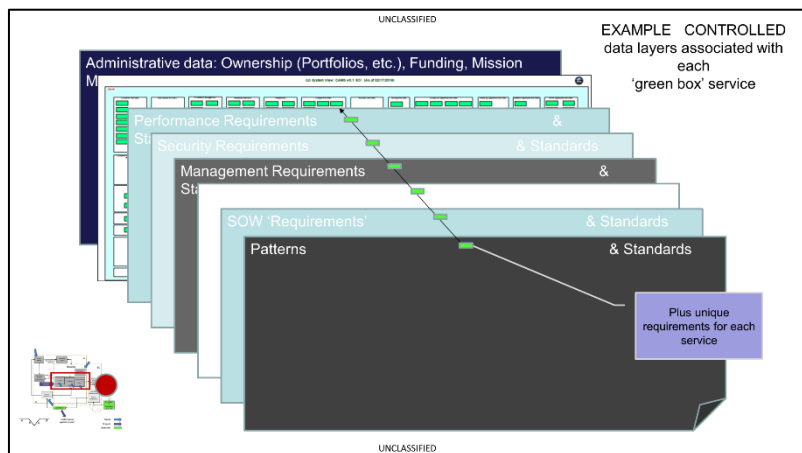


Figure B-6 - General Service Requirement Inheritance

business processes to accommodate new and evolving technologies. This will impact how to develop and operate discrete implementations within the enterprise. NGA may also modify its contracting process to align with a service-oriented acquisition approach.

Developing “as-a-service”:

- Enable stable APIs at fixed endpoints. Services are versioned or designed to be tolerant of varying client versions.
- Incorporate methods to understand service usage. Platform services (e.g., API gateway) should be leveraged to support the capture of usage data.
- Deploy and provision for high availability. Periodic outages are generally unacceptable.
- Adhere to a lifecycle to enable independent evolution of application, middleware, and infrastructure services independent of each other. New versions of services should maintain backwards compatibility to the maximum extent possible. Retirement decisions should be based on operational value to the mission and supportability.
- Deliver a consistent level of availability. Depending on the specifics of the service, scaling could be via automated (e.g. auto-scaling groups) or manually-initiated actions.
- Maintain test versions of APIs to allow others to develop and test against. Short interruptions or slow-downs are acceptable for test APIs.

Operating “as-a-service”:

- Consumers should interact directly with service providers and provide feedback on services, and their APIs, to the service provider.
- Interfaces are to be managed by the service provider. Solution engineers don’t require “requests for change” or Interface Control Documents (ICDs) to manage external transactions. Service should follow the standardized service development lifecycle (i.e., ITIL 3.0) and applicable standards.
- Solution engineers should regularly engage with users of GEOINT services and exchange ideas about how services can be improved; and, formulate upgrades to existing services or establish new service candidates.

APPENDIX C – MAPPING TRACEABILITY

Enterprise Systems Engineering & Integration (ESE&I) provides the enterprise integration function in order to maintain the enterprise baseline using metadata and service component relationships. ESE&I supports the governance process, collects and maintains data and artifacts to support configuration control via the control boards.

Mapping service relationships are key to maintaining baseline control. Mapping includes missions-to-capabilities-to-activities-to-services, as shown in Figure C-1.

ESE&I maintains the configuration control framework(s) in which the data and descriptions of the enterprise components (e.g., services, mission, organizations) are stored, as depicted. The engineering integration organization (i.e., ESE&I) maintains the enterprise baseline and handles the technical aspects of governance and decision-making for the NGA enterprise.

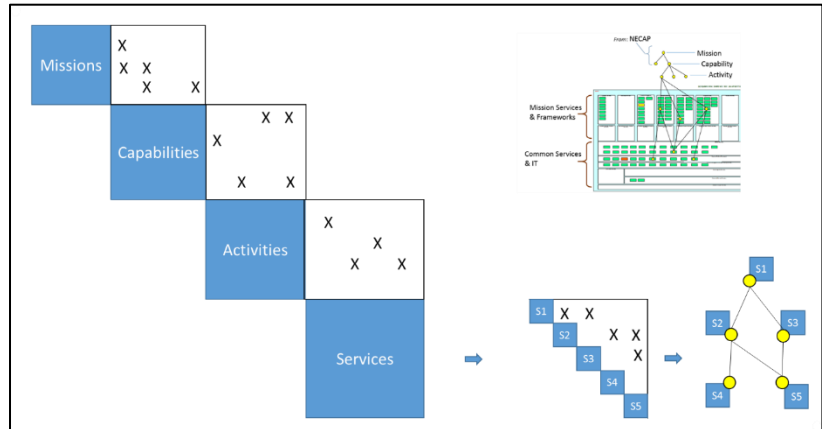


Figure C-1 - Missions Mapped to Services

The mission owner applies the mission (i.e., definition) while the PM and ESE&I apply the corresponding capabilities-to-activities-to-services framework. In cases where multiple missions use the same activities or services, ESE&I can arbitrate by maintaining and coordinating the technical documentation to support configuration control. ESE&I supports the enterprise governance process by maintaining data and artifacts to support configuration control at the service and program levels.

In order to trace costs and perform enterprise level trades, metadata is associated with service development and delivery, as shown in Figure C-2.

Service delivery and capability milestone dates (i.e., IOC, FOC) feed the schedule so that the enterprise schedule reflects the “actual” results from the service’s enterprise deployments. This allows mission capability timelines to be forecasted from development and capability-build schedules.

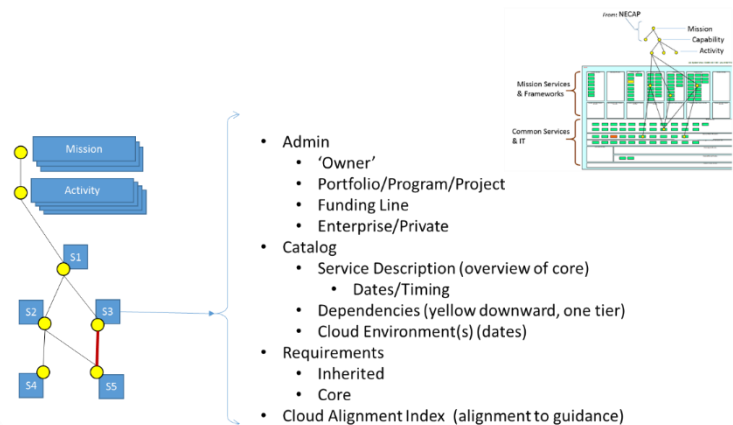


Figure C-2 - Metadata Associated with a Service

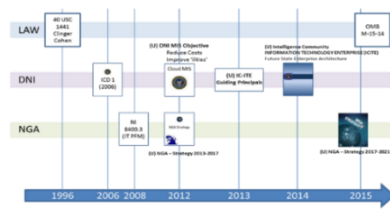
Using documented mission priorities as criteria, weights can be assigned to service capabilities, as shown in Figure C-3. These weights are reflected as values, based on capability growth and comply with the OMB requirements for value-driven decision-making.

Figure C-4 shows the schedule, capability roadmap and value roadmap used to manage service decisions within the NGA enterprise.

Figure C-5 illustrates mapping between the data matrices with the responsible owners. The diagram shows the trace from guidance-to-capability-to-budget allocation.

Services are mapped according to the JARM and DoDAF architecture frameworks (e.g., GEA CV-6/7, and ESL/TST).

Guiding Documentation



Data Structures

Values	.1	.3	.5	.2
Missions or Capabilities				

Figure C-3 - Value to Missions or Capabilities

Development schedule

Capability Roadmap

Value Roadmap

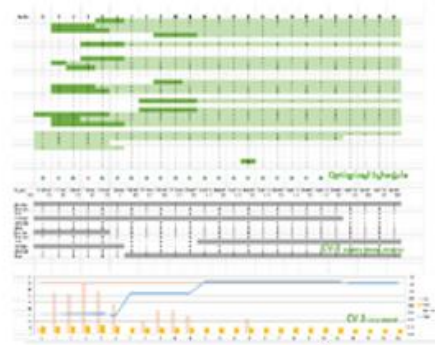


Figure C-4 - Capability and Value Roadmaps

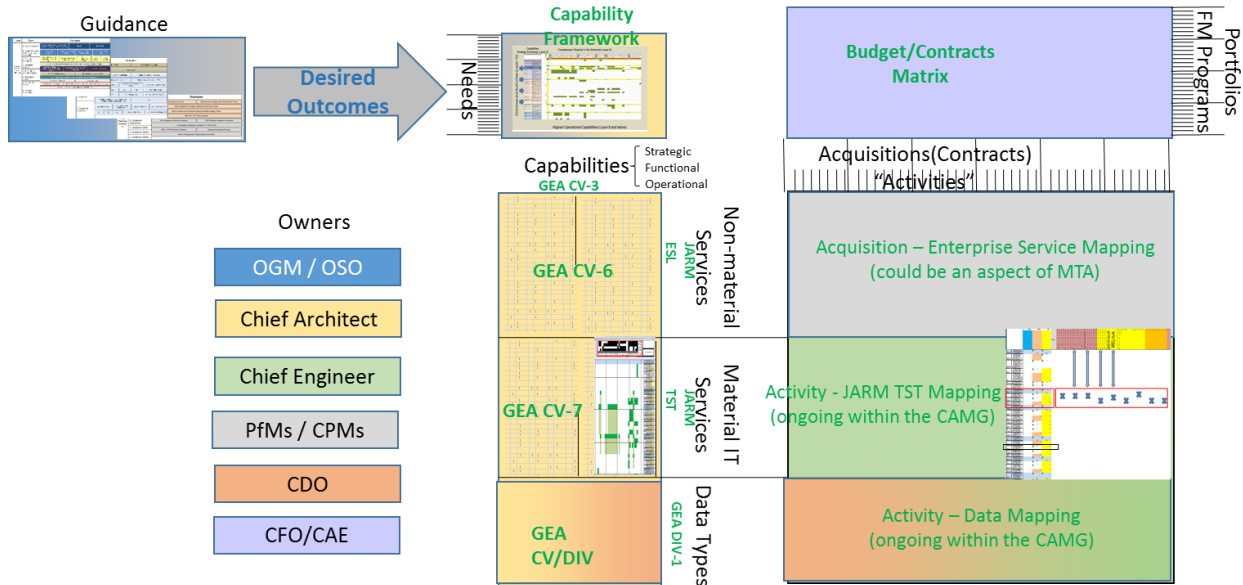


Figure C-5 - Mapping Services to Capabilities using Architecture Frameworks

APPENDIX D - SOFTWARE LICENSE DEFINITIONS

License Agreement Types	
Enterprise License Agreement (ELA) or Enterprise Agreement (EA)	Volume license type permitting product-use rights across all of a customer's computers and typically including upgrade rights.
End User License Agreement (EULA)	Similar to a software license agreement, a EULA contains terms and conditions for software use, but because it often lacks a license metric, it is typically insufficient as a proof of license. This is the software vendor's standard license agreement (favors the vendor, not the consumer)
Service-Level Agreement (SLA)	A contract describing the scope and measurements of work expected from a service provider to a customer throughout the term of the contract.
Software License Agreement (SLA)	Terms and conditions for rights to use a software product that is copyrighted intellectual property owned by the publisher
Master License Agreement	A type of volume license with deep product discounts for high-volume software purchasers
Corporate License Agreement (CLA)	A contract that features pricing per-user and per-device license metrics, used by Novell.
Unlimited License Agreement (ULA)	A contract that features (with Oracle only) allow unlimited deployment of specific products (CLIN 1 per IC-wide contract) and at the end of the contract each IC agency will true-up licenses on a per processor basis.
License Scope	
Scope	Software products and services applicable under a software license agreement.
Restrictive Use	Part of license agreement specifying what the licensee may or may not do with the software product. Under copyright law, all rights are reserved to the licensor except those specifically granted to the licensee, can also be program specific.
Unlimited Use	A contract permitting product-use rights to the entire organization
Third Party Use	Non-principal participant with responsibilities in a project, contract or some other agreement or event.
License Use	
Product Use Right (PUR)	Terms and conditions in a license agreement specifying how users can operate a product, and under what limitations, without breach of intellectual property, and thus of copyright law.
Named-User License	A contract constraining product-use rights to pre-designated individual users who access the product on multiple computers.
Device-Based License	A contract permitting product-use rights to individual devices, including, but not limited to, processors, servers and workstations. Any number of users may access and use the software on the allowed devices.
Processor-Based License	A contract pricing server software access by individual processors within the servers. Also called CPU based. (e.g., Oracle)
MAC-based License	The software license will only work on a specific MAC address.
Secondary-Use Right	Permission granted in a software license agreement or end user license agreement to install a second copy of a licensed software product on a second device, typically a portable computer and for the exclusive use of the licensee. Also known as home-use right, portable-use right.
Floating License	A contract permitting software transfer at any time between users and/or computers, though typically requiring de-installation from an asset at the time of installation on another asset.

Annex K: Cloud Business Guidance

	Licenses are, under copyright law, transferable from user to user and device to device unless expressly prohibited in a software license agreement. This type of license is often associated with named-user licenses.
Concurrent-Use License	A contract in which pricing of product-use rights is based on the maximum number of computers or users simultaneously accessing the software. The enforcement of compliance with this type of license requires metering.
True-Up License	Additional license purchased to reach compliance with a software license agreement through payment to compensate for product usage exceeding product-use rights or due to changes in licensing model by vendor (e.g., from core-based to processor-based licenses).
Capacity-Based License	Sets pricing for product-use rights based on hardware processing power or speed, e.g., hard-drive or central processing unit (CPU) size or speed.
Dual Operations Use	Use of the software licenses as per the specified time period in the contract's terms and conditions in which the same licenses can be used in operations in an existing NGA datacenter and in cloud test environment (Test VPC) in support of the program's cloud integration prior to going operational in the cloud.
License Types	
Perpetual License	Product-use rights for an unlimited duration, usually does not accommodate maintenance upgrades.
Term/Subscription License	Time-limited, non-perpetual contract constraining and product-use rights by a specified termination or renewal date, at which point the license must be renewed or the software removed.
Subscription Service (e.g., Cloud)	The license is provided via a vendor's cloud service through a monthly/annual subscription. When you quit paying the subscription fee, rights to use the software are lost. (Examples, Adobe Creative Cloud, Microsoft 365, etc.)
Temporary License	A set time limit for license use as agreed to by vendor and customer.
Evaluation License	Used to evaluate new products through Holistic, Alpha/Beta testing, or early adopter program. Post release evaluation to determine capabilities and/or security compliance at no cost and w/o an obligation to acquire products.
Developmental License	Can evaluate a new or improved software capability from a vendor for an agreed upon period in a test or lab environment.
No-Cost License	Provided at no cost to customer for such uses as training and testing in test environments.
Unlimited License	Product-use rights to the entire organization
Site License	Contract constraining product-use rights to a physical or logical location(s)
Volume License	Contract pricing multiple licenses at a discount, determined primarily by volume purchased, and secondarily by product type, license duration and other variables and license metrics
Client-Access License (CAL)	Use of software on a server, in which use rights are typically priced per individual server core, networked client device, user or seat. Commonly associated with rights for Microsoft server software.